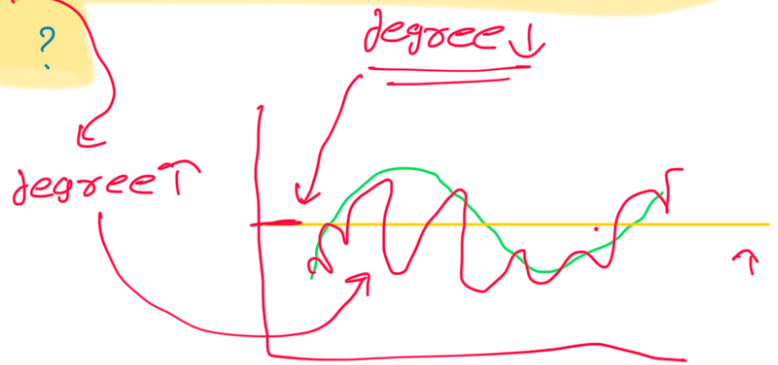


* Revision:

① Polynomial Reg.:



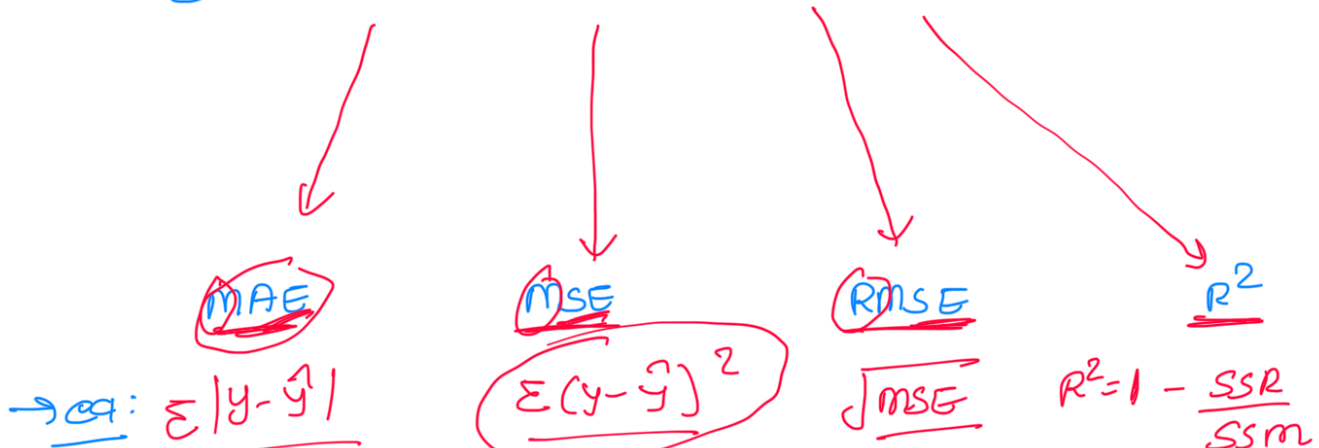
- When? →
- How? → $\otimes$ → $x^2 \ x^3 \ x^4$: degree
- How to train poly. Reg.? : $y = m_1x + m_2x^2 + m_3x^3 + c$
 $y = m_1x_1 + m_2x_2 + m_3x^3 + c$
- Hyperparameters?
- When overfitting & when Underfitting in polynomial Reg.?



Inter & dep. variable

② Regression metric:

① which Reg. metric we have studied?



n n n
 → Benefits: Same unit, unit d, same, Range?
 → Drawbacks: Robust out, outliers? affected by n , $[-\infty, \infty]$

⇒ Why we do $\frac{1}{n}$ (mean or Avg) of error?

total error $\propto$ no. of data points

$$\frac{\text{train } 100}{100} \gg \frac{\text{test } 20}{20}$$

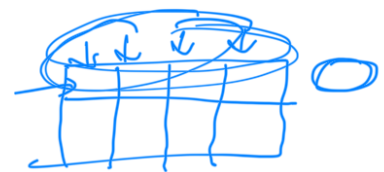
⇒ Student Research Question:

Output of MAE & RMSE will be same or not?
(In which case it will be same).

* Today's class:

① Adjusted R^2 : (R squared)

$$\text{Adj } R^2 = 1 - \frac{(1 - R^2)(n-1)}{n-k-1}$$



data point (row) 100
1000

n = no. of observation

k = no. of predictors (variables) $\rightarrow$

R^2 = og R^2 score

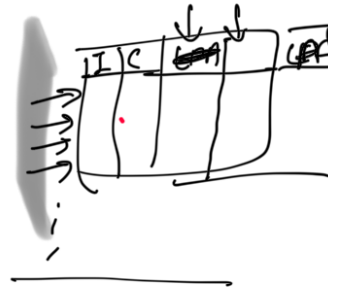
$\Rightarrow$ why we Reg. adj. R^2 we already have R^2 ?

Experiment

$R^2 = 0.8$
 $n = 10$
 $k = 2$

added two new features

$R^2 = 0.8$
 $n = 10$
 $k = 4$



$$\text{adj } R^2 = 1 - \frac{(1 - R^2)(n-1)}{(n-k-1)}$$

$$= 1 - \frac{(1 - 0.8)(9)}{7}$$

$$= 1 - 0.257$$

$$= 0.743$$

$$\text{adj } R^2 = 1 - \frac{(1 - R^2)(n-1)}{(n-k-1)}$$

$$= 1 - \frac{(1 - 0.8)(9)}{5}$$

$$= 1 - 0.36$$

$$= 0.64$$

Model 1: 1 feature(s) -> ['study hours']
Model 2: 2 feature(s) -> ['study hours', 'shoe_size']
Model 3: 3 feature(s) -> ['study hours', 'shoe_size', 'random_feature_1']
Model 4: 4 feature(s) -> ['study hours', 'shoe_size', 'random_feature_1', 'random_feature_2']
Model 5: 5 feature(s) -> ['study hours', 'shoe_size', 'random_feature_1', 'random_feature_2', 'random_feature_3']
Model 6: 6 feature(s) -> ['study hours', 'shoe_size', 'random_feature_1', 'random_feature_2', 'random_feature_3', 'random_feature_4']

All models were trained on the identical training rows.

training Error $\downarrow$
testing Error $\uparrow$

model	n_features	train_r2	train_adj_r2	test_r2	test_adj_r2	train_mae	test_mae	train_rmse	test_rmse
0 Model 1	1	0.7360	0.7334	0.7314	0.7251	6.8997	6.4896	8.5106	8.5373
1 Model 2	2	0.7361	0.7310	0.7276	0.7146	6.9024	6.5095	8.5078	8.5974
2 Model 3	3	0.7376	0.7298	0.7248	0.7046	6.8723	6.5889	8.4836	8.6414
3 Model 4	4	0.7400	0.7296	0.7186	0.6904	6.8477	6.6670	8.4460	8.7382

4	Model 5	5	0.7404	0.7273	0.7166	0.6803	6.8135	6.6732	8.4385	8.7688
5	Model 6	6	<u>0.7413</u>	<u>0.7255</u>	<u>0.7128</u>	<u>0.6675</u>	6.8054	6.7096	8.4239	8.8268

• R^2 Rewards adding features even useless ones. Adj R^2 It gives credit only when a new feature truly Improves the model performance.

→ Adj. R^2 penalizes unnecessary features.

→ Why training $R^2 \uparrow$ when we add unnecessary features?

→ overfitting (one more feature → one more dimension to overfit)

→ when to use Adj R^2 ?

→ when we want to compare model with diff. no. of features.

⇒ limitation of Adj R^2 :

- It only applies to linear Regression models.

(S.L.R., MLR, Poly. Reg.)

- Not useful for D.T., RF, xgboost...

→ Why? Parameters of model complexity.

$$\underline{y = mx + b}$$

$\alpha <$

$$y = m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4 + \epsilon$$

↑
more comp

